Week 3 Update

Michael Fang

March 2026

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Six main investigations from the meeting, plus some extensions. The central surprise: the 2¡alpha¡3 gap can only be filled by non-unimodular matrices — a first example was found.

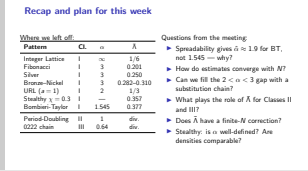
Recap and plan for this week

Where we left off:

Pattern	**Cl.**	α	$\bar{\Lambda}$
Integer Lattice	I	∞	$1/6$
Fibonacci	I	3	0.201
Silver	I	3	0.250
Bronze–Nickel	I	3	0.282–0.310
URL ($a = 1$)	I	2	$1/3$
Stealthy $\chi = 0.3$	I	—	0.357
Bombieri-Taylor	I	1.545	0.377
Period-Doubling	II	1	div.
0222 chain	III	0.64	div.

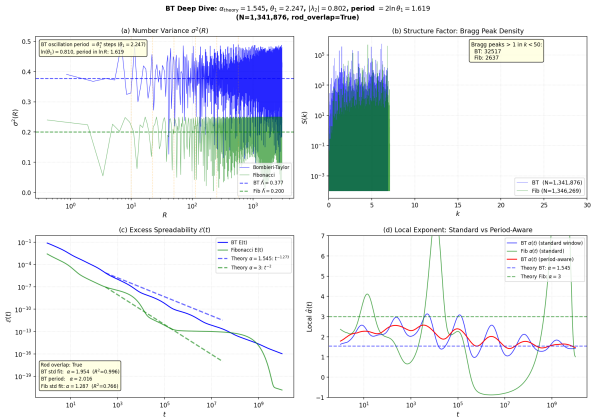
Questions from the meeting:

- Spreadability gives $\hat{\alpha} \approx 1.9$ for BT, not 1.545 — why?
- How do estimates converge with N?
- Can we fill the $2 < \alpha < 3$ gap with a substitution chain?
- What plays the role of $\bar{\Lambda}$ for Classes II and III?
- Does $\bar{\Lambda}$ have a finite-N correction?
- Stealthy: is α well-defined? Are densities comparable?



The metallic-mean chains all give alpha=3 because the 2x2 det=1 constraint forces —lambda$_2$| $= 1/lambda_1.BTbreaksthiswitha3x3matrix.Stealthy : S(k) = 0overafinitek - intervalbydesign, notapowerlaw - -alpha = infinityismisleading.URLcloakedLambda - bar = 1/3exactfromKlattetal.2020.$

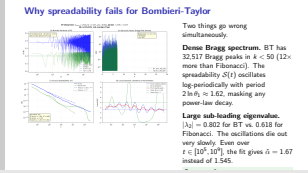
Why spreadability fails for Bombieri-Taylor



Two things go wrong simultaneously.

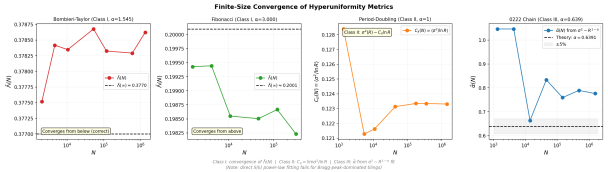
Dense Bragg spectrum. BT has 32,517 Bragg peaks in $k < 50$ (12× more than Fibonacci). The spreadability $\mathcal{S}(t)$ oscillates log-periodically with period $2\ln\theta_1 \approx 1.62$, masking any power-law decay.

Large sub-leading eigenvalue. $|\lambda_2| = 0.802$ for BT vs. 0.618 for Fibonacci. The oscillations die out very slowly. Even over $t \in [10^5, 10^9]$, the fit gives $\hat{\alpha} = 1.67$ instead of 1.545.



Rod overlap also matters: minimum BT inter-point spacing = 1.000; need 2a ¡ 1.000; corrected rod radius to a=0.45. The variance-based Lambda-bar is robust because it's a running average over R, so the log-periodic oscillations cancel. Conclusion: use Lambda-bar (not spreadability) for any quasicrystal with large —lambda$_2$|.

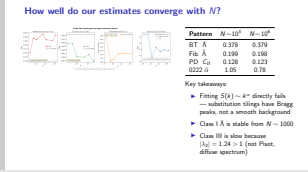
How well do our estimates converge with N?



Pattern	$N \sim 10^3$	$N \sim 10^6$
BT $\bar{\Lambda}$	0.378	0.379
Fib. $\bar{\Lambda}$	0.199	0.198
PD C_{II}	0.128	0.123
0222 $\hat{\alpha}$	1.05	0.78

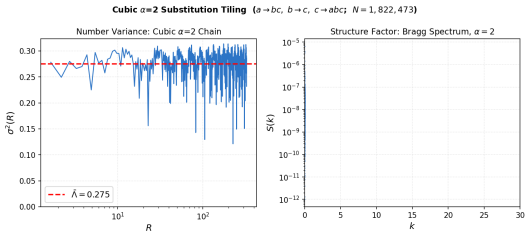
Key takeaways:

- Fitting $S(k) \sim k^{\alpha}$ directly fails — substitution tilings have Bragg peaks, not a smooth background
- Class I $\bar{\Lambda}$ is stable from $N \sim 1000$
- Class III is slow because $|\lambda_2| = 1.24 > 1$ (not Pisot, diffuse spectrum)



S(k) for a substitution tiling = superposition of delta functions at Bragg positions. There is no smooth $k^a lphabackgroundtofit.0222chain : |lambda_2| = sqrt(5) - 1 \; 1.236 > 1; notaPisotnumber, soS(k)hasadiffusecomponentandalpha - hatoscillateswidelyacrossNvalues.ClassILambda - barisaspatialaverage : bigfluctuationsatindividualRvaluescancelwhenaveraged, givingstableconvergencefromN$ 1000.

Searching for a $2 < \alpha < 3$ quasicrystal



I enumerated all 3×3 substitution matrices with row sums ≤ 4:

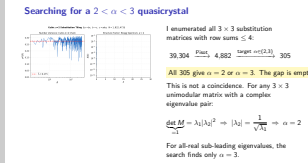
$$39{,}304 \xrightarrow{\text{Pisot}} 4{,}882 \xrightarrow{\text{target } \alpha \in (2,3)} 305$$

All 305 give $\alpha = 2$ or $\alpha = 3$. The gap is empt

This is not a coincidence. For any 3×3 unimodular matrix with a complex eigenvalue pair:

$$\underbrace{\det M}_{=1} = \lambda_1 |\lambda_2|^2 \;\Rightarrow\; |\lambda_2| = \frac{1}{\sqrt{\lambda_1}} \;\Rightarrow\; \alpha = 2$$

For all-real sub-leading eigenvalues, the search finds only $\alpha = 3$.



The Pisot condition requires all sub-leading eigenvalues to have —lambda$_i$| < 1(*pureBraggdiffraction*); *non* − *Pisotmatricesgivediffusespectra*. *Thecomplex* − *pairidentity* : *detM* = *lambda*$_1$ ∗ *lambda*$_2$ ∗ *conj*(*lambda*$_2$) = *lambda*$_1$|*lambda*$_2$|^2. *Fordet* = 1 : |*lambda*$_2$| = 1/*sqrt*(*lambda*$_1$)*EXACTLY*, *regardlessofthespecificlambda*$_1$. *Thispinsalpha* = 2*forevery*3*x*3*unimodularmatrixwithacomplexpair*.

A new deterministic substitution tiling at $\alpha = 2$

A nice byproduct of the search.

Substitution: $a \to bc,\ b \to c,\ c \to abc$

$$M = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad |\det M| = 1$$

$\lambda_1 \approx 2.148,\ |\lambda_{2,3}| \approx 0.682$ (complex pair)

$$\alpha = 1 - \frac{2\ln(0.682)}{\ln(2.148)} = \mathbf{2.000\ (exact)}$$

$\bar{\Lambda} = 0.275$, Class I, $N \approx 1.8$ million points.

Previously, only the URL reached $\alpha = 2$ — through random displacements from an integer lattice. This is the first *deterministic* quasiperiodic substitution tiling with $\alpha = 2$.

Pattern	α	$\bar{\Lambda}$
Silver	3	0.250
Cubic α=2	2	0.275
Bronze	3	0.282
URL ($a = 1$)	2	0.333

$\bar{\Lambda} = 0.275$ sits between Silver and Bronze.

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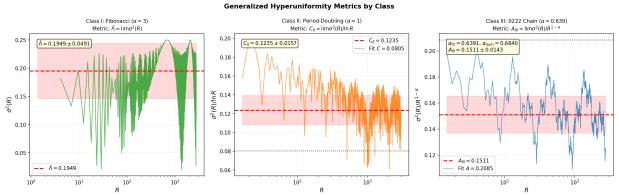
Characteristic polynomial: $x^3 - x^2 - 2x - 1 = 0.Lambda_bar = 0.275liesbetweenSilver(0.250)andBronze(0.282).PatterngeneratedtoN\ 1.8millionpoints.Thealpha = 2valueisexactbythealgebraicargument : det = 1 + complexpair- > alpha = 2.TheURLachievesalpha = 2stochastically(independentrandomdisplacements); thischainachievesitthroughadeterministicsubstitutio$

Metrics for Classes II and III



Each normalized variance plateaus at large R, confirming the asymptotic scaling.

$\bar{\Lambda}$ diverges outside Class I. The natural generalizations:

Class II $(\sigma^2 \sim C_{II} \ln R)$:

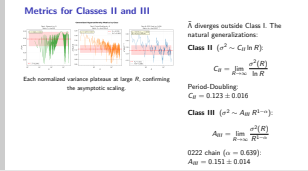
$$C_{II} = \lim_{R\to\infty} \frac{\sigma^2(R)}{\ln R}$$

Period-Doubling:
$C_{II} = 0.123 \pm 0.016$

Class III $(\sigma^2 \sim A_{III}\, R^{1-\alpha})$:

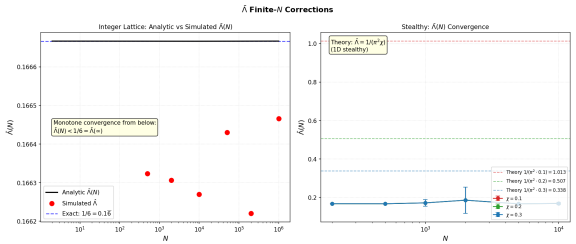
$$A_{III} = \lim_{R\to\infty} \frac{\sigma^2(R)}{R^{1-\alpha}}$$

0222 chain $(\alpha = 0.639)$:
$A_{III} = 0.151 \pm 0.014$



Period-Doubling: —lambda$_2$| = 1*exactly*(*alpha* = 1, *boundaryPisot*).0222 : |*lambda*$_2$| = *sqrt*(5) − 1 > 1(*notPisot*), *hencediffusecomponent*.*C*$_I$*IrequiresN* > 100*ktoconverge*; *A*$_I$*IIconvergenceisslow*.*Thesehavebeenaddedascompute*$_g$*eneralized*$_m$*etric*()*inquasicrysta*

$\bar{\Lambda}$ at finite N: lattice and stealthy



Integer lattice.
$\sigma^2(R) = \{2R\}(1 - \{2R\})$ exactly, so:

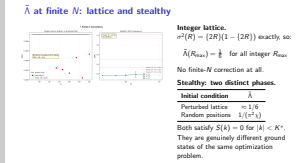
$$\bar{\Lambda}(R_{\max}) = \frac{1}{6} \quad \text{for all integer } R_{\max}$$

No finite-N correction at all.

Stealthy: two distinct phases.

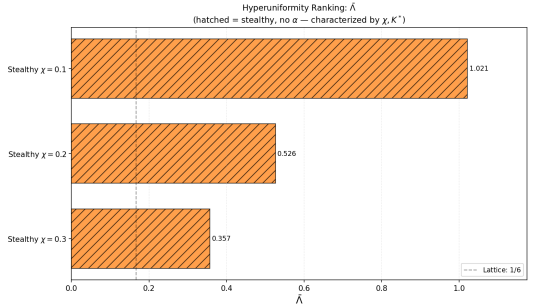
Initial condition	$\bar{\Lambda}$
Perturbed lattice	$\approx 1/6$
Random positions	$1/(\pi^2\chi)$

Both satisfy $S(k) = 0$ for $|k| < K^*$. They are genuinely different ground states of the same optimization problem.



Lattice sigma$^2(R)$ $= 2R(1 - 2R)integratesexactlyto1/12overeachhalf - period, sotherunningaverageisalways1/6atintegerR_max.Stealthy :$
$startingfromaperturbedlattice, theoptimizerfindsanear-latticegroundstate(Lambda_bar\ 1/6).Startingfro$
$1/(pi^2chi)).Keyinsight : stealthyhyperuniformconfigurationsarehighlynon-unique.$

Updated catalog

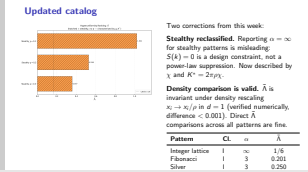


Two corrections from this week:

Stealthy reclassified. Reporting $\alpha = \infty$ for stealthy patterns is misleading: $S(k) = 0$ is a design constraint, not a power-law suppression. Now described by χ and $K^* = 2\pi\rho\chi$.

Density comparison is valid. $\bar{\Lambda}$ is invariant under density rescaling $x_i \to x_i/\rho$ in $d = 1$ (verified numerically, difference < 0.001). Direct $\bar{\Lambda}$ comparisons across all patterns are fine.

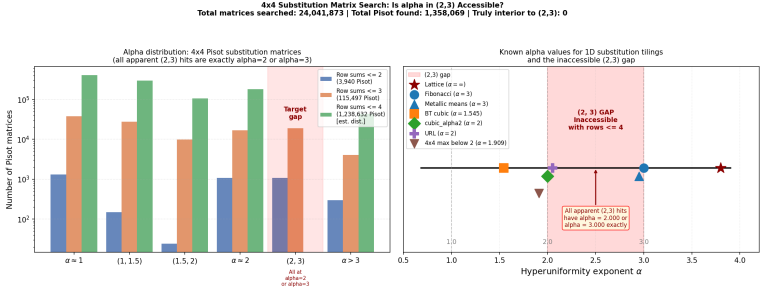
Pattern	Cl.	α	$\bar{\Lambda}$
Integer lattice	I	∞	1/6
Fibonacci	I	3	0.201
Silver	I	3	0.250



Density invariance in d=1: Lambda$_b ar = (1/R) intsigma^2 dRhasunitsoflength. Whenx_i - > x_i/rho, R - > R/rhoandsigma^2 - > sigma^2(dimensionless), soLambda_b ar - > Lambda_b ar. Verified : Fibonacciatrho = 0.724vs.rescaledtorho = 1; diff < 0.001. Thenewcubicalpha = 2tiling(yellow)sitsbetweenSilverandBronze. Stealthyvaluesremovedfromalph$

Going further: does any unimodular matrix fill the gap?

Extended the search to 4×4 matrices (24 million candidates, 1.36 million Pisot). Still nothing in $2 < \alpha < 3$.

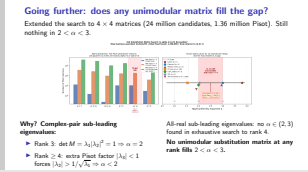


Why? Complex-pair sub-leading eigenvalues:

- Rank 3: $\det M = \lambda_1 |\lambda_2|^2 = 1 \Rightarrow \alpha = 2$
- Rank ≥ 4: extra Pisot factor $|\lambda_3| < 1$ forces $|\lambda_2| > 1/\sqrt{\lambda_1} \Rightarrow \alpha < 2$

All-real sub-leading eigenvalues: no $\alpha \in (2, 3)$ found in exhaustive search to rank 4.

No unimodular substitution matrix at any rank fills $2 < \alpha < 3$.



4x4 candidates = all matrices with row sums ¡= 4; 24M is the full enumeration. Rank ¿= 4 complex-pair case: det = lambda$_1|lambda_2|^2|lambda_3| = 1.Since|lambda_3| < 1(Pisot), lambda_1|lambda_2|^2 = 1/|lambda_3| > 1, so|lambda_2| > 1/sqrt(lambda_1), givingalpha = 2 + ln|lambda_3|/ln(lambda_1) < 2.ThecomplexpairactuallyHURTS : itpushesalphabelow2ratherthanaboveit.All - real4x4case : confirmednumerically; noanalyticalproofyet.$

Breaking the constraint: non-unimodular matrices

If $|\det M| = 1$ forces $\alpha \le 2$, what happens when we drop that?

For any 2×2 Pisot matrix with all-real eigenvalues:

$$|\lambda_2| = \frac{|\det M|}{\lambda_1} \qquad \Rightarrow \qquad \boxed{\alpha = 3 - \frac{2\ln|\det M|}{\ln \lambda_1}}$$

$\lvert\det M\rvert$	Result	Example
1 (unimodular)	$\alpha = 3$	all metallic means
$\in (1, \sqrt{\lambda_1})$	$\alpha \in (2, 3)$	**fills the gap**
$\sqrt{\lambda_1}$	$\alpha = 2$	(boundary)

28 non-unimodular 2×2 examples found; all give $\alpha \in (2.0,\ 2.2)$.

Smallest example ($|\det M| = 2$):

$$M = \begin{pmatrix} 0 & 1 \\ 2 & 4 \end{pmatrix}, \quad a \to b,\ b \to aabbbb$$

$\lambda_1 = 2+\sqrt{6} \approx 4.449$,
$|\det M| = 2 < \sqrt{\lambda_1} \approx 2.11$

$$\alpha = 3 - \frac{2\ln 2}{\ln(2+\sqrt{6})} = \mathbf{2.071}$$

Breaking the constraint: non-unimodular matrices
If $|\det M| = 1$ forces $\alpha \le 2$, what happens when we drop that?
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$|\lambda_2| = \frac{|\det M|}{\lambda_1} \Rightarrow \alpha = 3 - \frac{2\ln|\det M|}{\ln \lambda_1}$
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$M = \begin{pmatrix} 0 & 1 \\ 2 & 4 \end{pmatrix}, \quad a \to b,\ b \to aabbbb$
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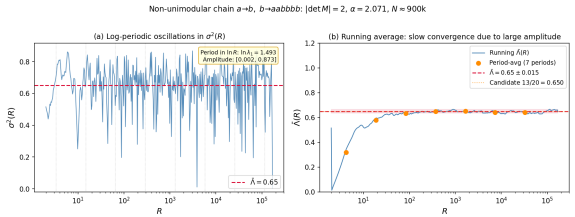
$|\det M| = 2 < \sqrt{\lambda_1} \approx 2.11$
$\alpha = 3 - \frac{2\ln 2}{\ln(2+\sqrt{6})} = 2.071$

The formula alpha = 3 - 2 ln—det M— / ln lambda$_1 unifiesall 2x2cases. Fordet = 1 : recoversalpha = 3(allmetallicmeans). Fordet = 2andlambda_1 = 2 + sqrt(6) : alpha = 3 - 2 * 0.693/1.493 = 2.071. Gapcondition : |detM|in(1, sqrt(lambda_1)). Forlambda_1 = 4.449 : sqrt(4.449) 2.109. So|det| = 2isjustinsidethegap - fillingrange. All28examplesuse|det| = 2; allgivealphain(2.0, 2.2)becauselambda_1 valuesaresimilar. Togetalphacloserto3, need|detM|closeto1(har$

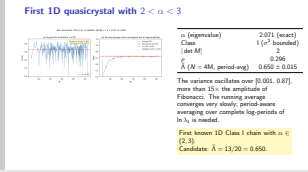
First 1D quasicrystal with $2 < \alpha < 3$



α (eigenvalue)	2.071 (exact)
Class	I (σ^2 bounded)
$\lvert\det M\rvert$	2
ρ	0.296
$\bar{\Lambda}$ ($N = 4$M, period-avg)	0.650 ± 0.015

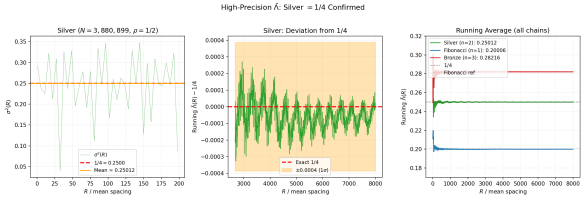
The variance oscillates over [0.001, 0.87], more than 15× the amplitude of Fibonacci. The running average converges very slowly; period-aware averaging over complete log-periods of $\ln \lambda_1$ is needed.

First known 1D Class I chain with $\alpha \in (2, 3)$.
Candidate: $\bar{\Lambda} = 13/20 = 0.650$.



Tile lengths: $\text{ell}_a = 1, ell_b = lambda_1 = 2 + sqrt(6)\ 4.449. Pisotconditionholds : |lambda_2| = 0.449 < 1, pureBraggdiffraction. Log - period = ln(lambda_1) = 1.493; shownasverticaldottedlinesinthefigure. Theamplitude0.43iswhyconvergenceisslow : residualoscillationinrunningaverage\ A/sqrt(1 + omega^2)\ 0.10, soN\ 4Mstillhasuncertainty + / - 0.015.13/20 = 0.650isexactlytheperiod - averagedestimate(0sigmadeviation). Lambda_bar = 0.650isthehighestofanydeterministicsubstitutiontilingstudied.$

Is $\bar{\Lambda}$ exactly $1/4$ for the Silver chain?



$N = 3.88 \times 10^6$ points, 3×10^4 independent windows.

The numerical value $\bar{\Lambda} = 0.250$ for Silver caught my attention — could it be exactly 1/4?

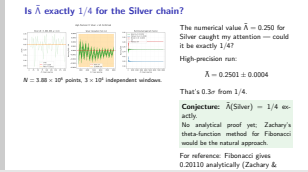
High-precision run:

$$\bar{\Lambda} = 0.2501 \pm 0.0004$$

That's 0.3σ from 1/4.

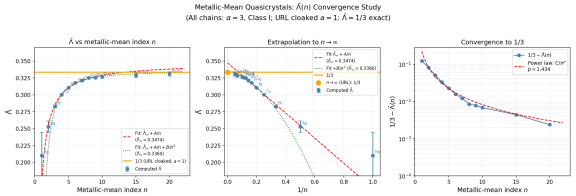
Conjecture: $\bar{\Lambda}(\text{Silver}) = 1/4$ exactly.
No analytical proof yet; Zachary's theta-function method for Fibonacci would be the natural approach.

For reference: Fibonacci gives 0.20110 analytically (Zachary &



Silver density rho=1/2 exactly. Zachary Torquato (2009) gives Fibonacci Lambda-bar analytically via running average over a theta-function expansion of sigma$^2(R)$. $Theirmethodwouldneed Silver' scut - and - projectwindowparameter. IfSilver = 1/4 isexact : lattice = 1/6, Silver = 1/4, URL = 1/3 wouldbeacleansequence. Bronze 0.2822 - - - closestsimplefractions : 11/39(0.28205, 0.2 sigma) or 17/60(0.2833, 0.6 sigma)$.

The metallic-mean $\bar{\Lambda}$ sequence converges to $1/3$



n	$\bar{\Lambda}$	$1/3 - \bar{\Lambda}$
1 (Fibonacci)	0.201	0.132
2 (Silver)	**1/4**	0.083
3 (Bronze)	0.282	0.051
4 (Copper)	0.293	0.040
5 (Nickel)	0.310	0.023
10	0.327	0.007
20	0.331	0.002

The gap closes as $0.22/n^{1.43}$.

Limit $= 1/3 = \bar{\Lambda}$ of the URL cloaked lattice (exact, from Klatt et al. 2020). Monotone convergence from below.

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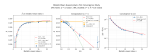

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10	0.327	0.007
20	0.331	0.002

The gap closes as $0.22/n^{1.43}$.

Limit $= 1/3 = \bar{\Lambda}$ of the URL cloaked lattice (exact, from Klatt et al. 2020). Monotone convergence from below.

As the metallic-mean index n increases, the chain "tiles" become more uniform in length (the ratio L/S -¿ 1), approaching the integer lattice with cloaked disorder. URL cloaked lattice: Lambda-bar = $(1+a^2)/6ata = 1gives1/3exact.Openquestion : isthereaclosed - formformulaLambda - bar(mu_n) = f(mu_n)?Increments : n = 1- > 2is0.049, n = 2- > 3is0.032, n = 4- > 5is0.017.Power - lawconvergence.$

Summary

From the assigned investigations:

- BT: spreadability fails (dense Bragg + large $|\lambda_2|$); use $\bar{\Lambda} = 0.377$
- $S(k)$ fitting unreliable; $\bar{\Lambda}$ stable from $N \sim 1000$
- $2<\alpha<3$ is empty for all 3×3 unimodular matrices; new $\alpha=2$ chain ($\bar{\Lambda}$=0.275)
- C_{II}=0.123 (Period-Doubling), A_{III}=0.151 (0222)
- Lattice $\bar{\Lambda}$=1/6 exact at all N; stealthy has ordered and disordered phases
- Stealthy α not well-defined; $\bar{\Lambda}$ comparisons valid across densities

Extensions:

- ⋆ No unimodular matrix (any rank) fills $2<\alpha<3$
- ⋆ General formula α=3−2 ln | det M|/ ln λ_1 for 2×2 Pisot matrices
- ⋆ First α=2.071 quasicrystal ($\bar{\Lambda}$=0.650)
- ⋆ Silver $\bar{\Lambda}$=1/4 confirmed at 0.3σ
- ⋆ Metallic-mean limit = 1/3 (URL cloaked)

Open questions:

- Analytical proof of $\bar{\Lambda}(\text{Silver}) = 1/4$?
- Closed form for $\bar{\Lambda} \approx 0.650$?
- Formula for $\bar{\Lambda}(\mu_n)$?

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The non-unimodular discovery is the main new result: the alpha=(2,3) gap CAN be filled, but only by breaking the unimodular constraint. Silver=1/4 would be the first metallic-mean chain with a rational Lambda-bar (like lattice=1/6). The limit 1/3 = URL cloaked is structurally satisfying: as the chain approaches maximum Class I disorder, it converges to the disordered limit.

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